RUBRIC COMPONENT #7 (4 MARKS)	OFFICIAL CRITERIA: Nested/correlated/recursive queries, JOIN SQL queries

Nested, Correlated, Recursive & Multi-Table Joins

Course: DBMS Mini Project (UE24CS395) • **Institution:** PES University (2026–2027)
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1. Complex Multi-Table JOIN (6 Relations)

```
SELECT c.case_id, c.title, o.full_name AS lead_officer, p.full_name AS suspect, cp.role_in_case, e.title AS exhibit
FROM cases c
JOIN officers o ON c.lead_officer_id = o.officer_id
JOIN case_people cp ON c.case_id = cp.case_id
JOIN people p ON cp.person_id = p.person_id
LEFT JOIN evidence e ON c.case_id = e.case_id
WHERE c.priority IN ('High', 'Critical') LIMIT 2;
-- OUTPUT:
-- CASE-2026-001 | Op Shrouded Horizon | Rajesh Sharma | Vikram Singh | Principal Target | ThinkPad X1 Carbon
```

2. Correlated Subquery (Identifying Multi-Case Subjects)

```
SELECT p.person_id, p.full_name, p.category,
    (SELECT COUNT(DISTINCT cp.case_id) FROM case_people cp WHERE cp.person_id = p.person_id) AS active_cases
FROM people p
WHERE (SELECT COUNT(DISTINCT cp2.case_id) FROM case_people cp2 WHERE cp2.person_id = p.person_id) >= 2;
-- OUTPUT:
-- PER-001 | Vikram Singh   | Subject | 2
-- PER-002 | Rohan Deshmukh | Subject | 2
```

3. Recursive Common Table Expression (CTE): Multi-Hop Criminal Network

```
WITH RECURSIVE SuspectNetwork AS (
    -- Anchor Member (Depth 0)
    SELECT cp.person_id AS source_id, cp.person_id AS target_id, cp.case_id, 0 AS depth
    FROM case_people cp WHERE cp.person_id = 'PER-001'
    UNION
    -- Recursive Member (Depth N+1)
    SELECT sn.source_id, next_cp.person_id, next_cp.case_id, sn.depth + 1
    FROM SuspectNetwork sn
    JOIN case_people curr_cp ON sn.target_id = curr_cp.person_id
    JOIN case_people next_cp ON curr_cp.case_id = next_cp.case_id
    WHERE next_cp.person_id != sn.target_id AND sn.depth < 2
)
SELECT DISTINCT p.person_id, p.full_name, sn.depth, sn.case_id
FROM SuspectNetwork sn JOIN people p ON sn.target_id = p.person_id ORDER BY sn.depth ASC;
-- OUTPUT:
-- PER-001 | Vikram Singh   | 0 (Root Subject)      | CASE-2026-001
-- PER-002 | Rohan Deshmukh | 1 (Direct Associate)  | CASE-2026-001
-- PER-007 | Meera Sen      | 2 (Indirect Contact)  | CASE-2026-002
```